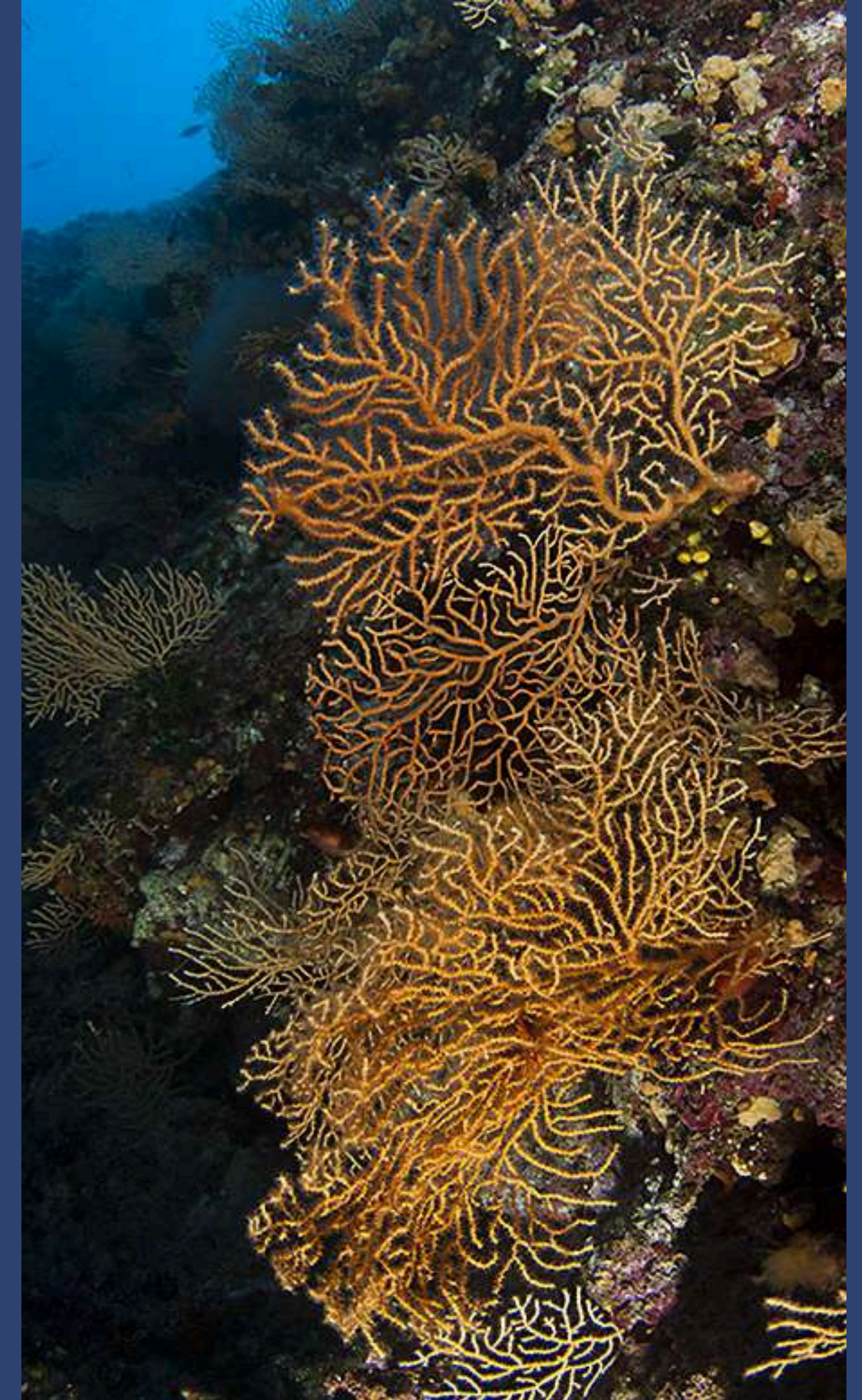




Life after Gorgonian's forest

*Andrea Calocero
Alessandro Mariani
Eleonora C. Savino*

*Nicole A. Ebana
Salvatore Piazza
Simone Miceli*



Gorgonacea lamouroux

- Fan corals, called **gorgonians**, are marine animals that form large, flattened, branched colonies
- These structures are formed by numerous small polyps, similar to anemones and jellyfish, which build a central horny skeleton called a **gorgonin**
- *Eunicella cavolinii* is a widely distributed gorgonian in the Mediterranean Sea, found from 15 to 200 metres depth. Its colonies can reach up to 60 cm in height

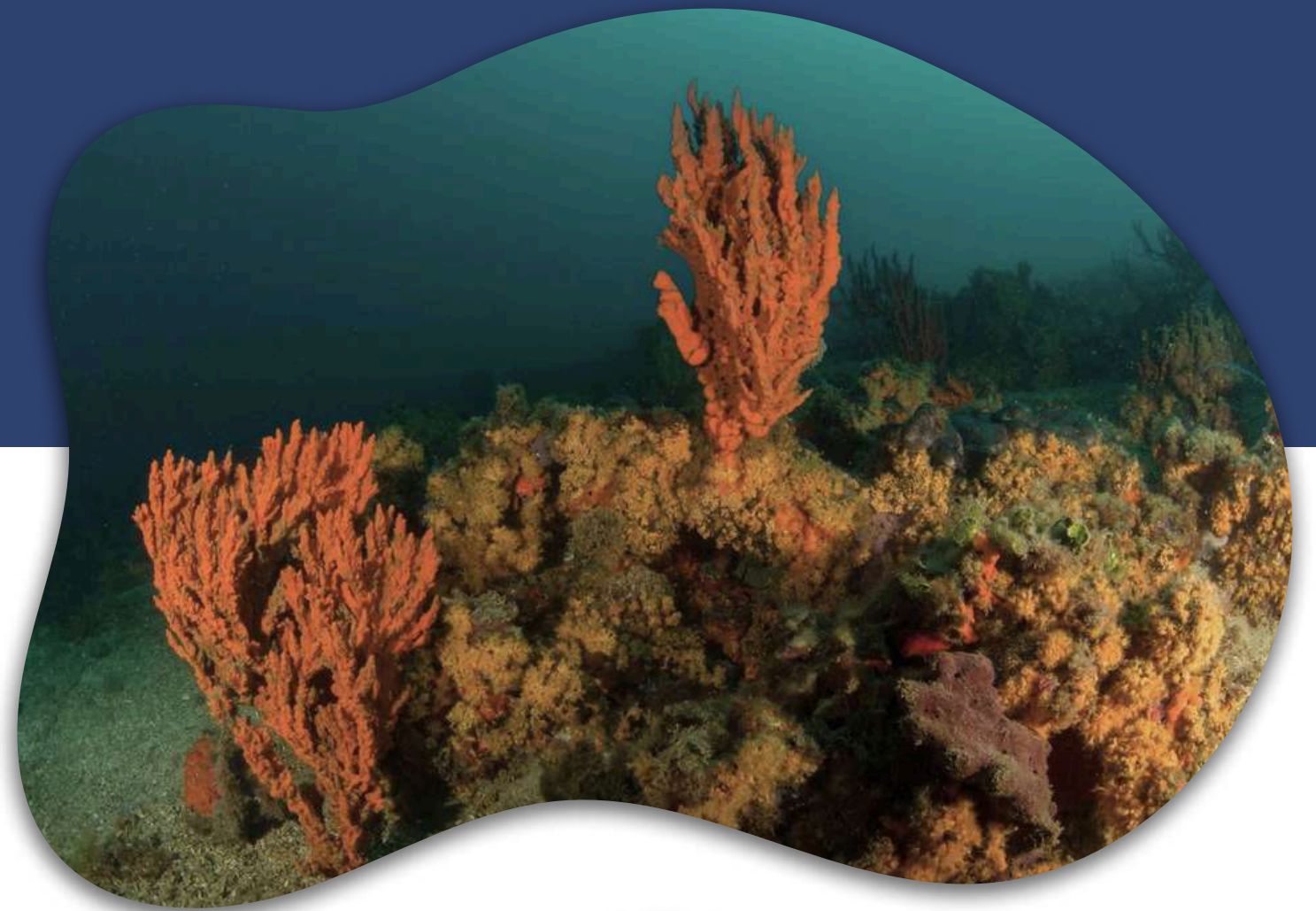


E. cavolinii

Ecological functions

Gorgonians perform an important ecological function:

- Create 3D structures, called “**animal forests**”
- Increase the complexity of the substrate and the biodiversity of the area
- Create new ecological niches.



THREATS

to the ecological balance



Anthropogenic events

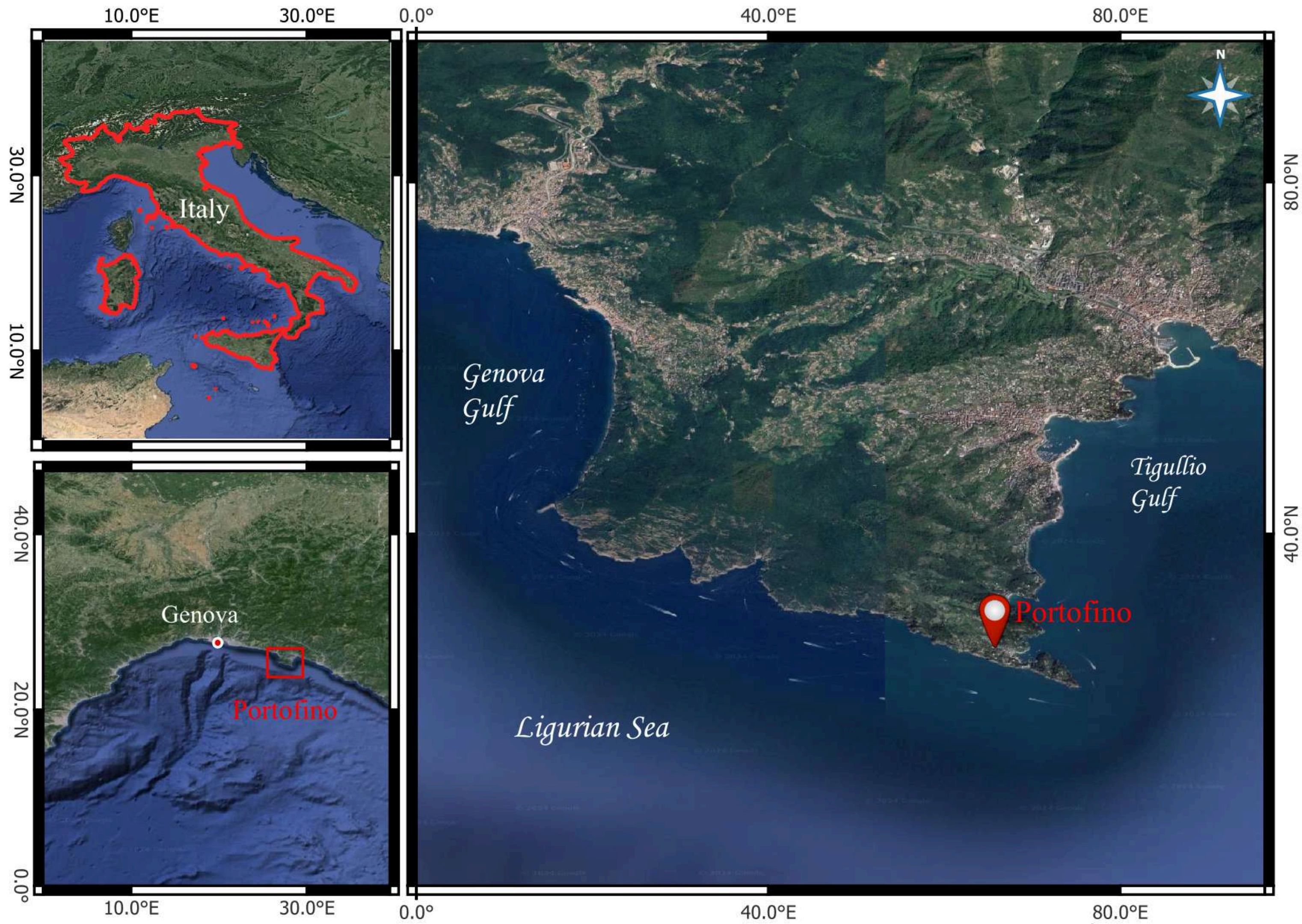
- Anchorage
- Scuba diving
- Recreational and commercial fishing
- Fishing gear and direct impacts with lost longlines

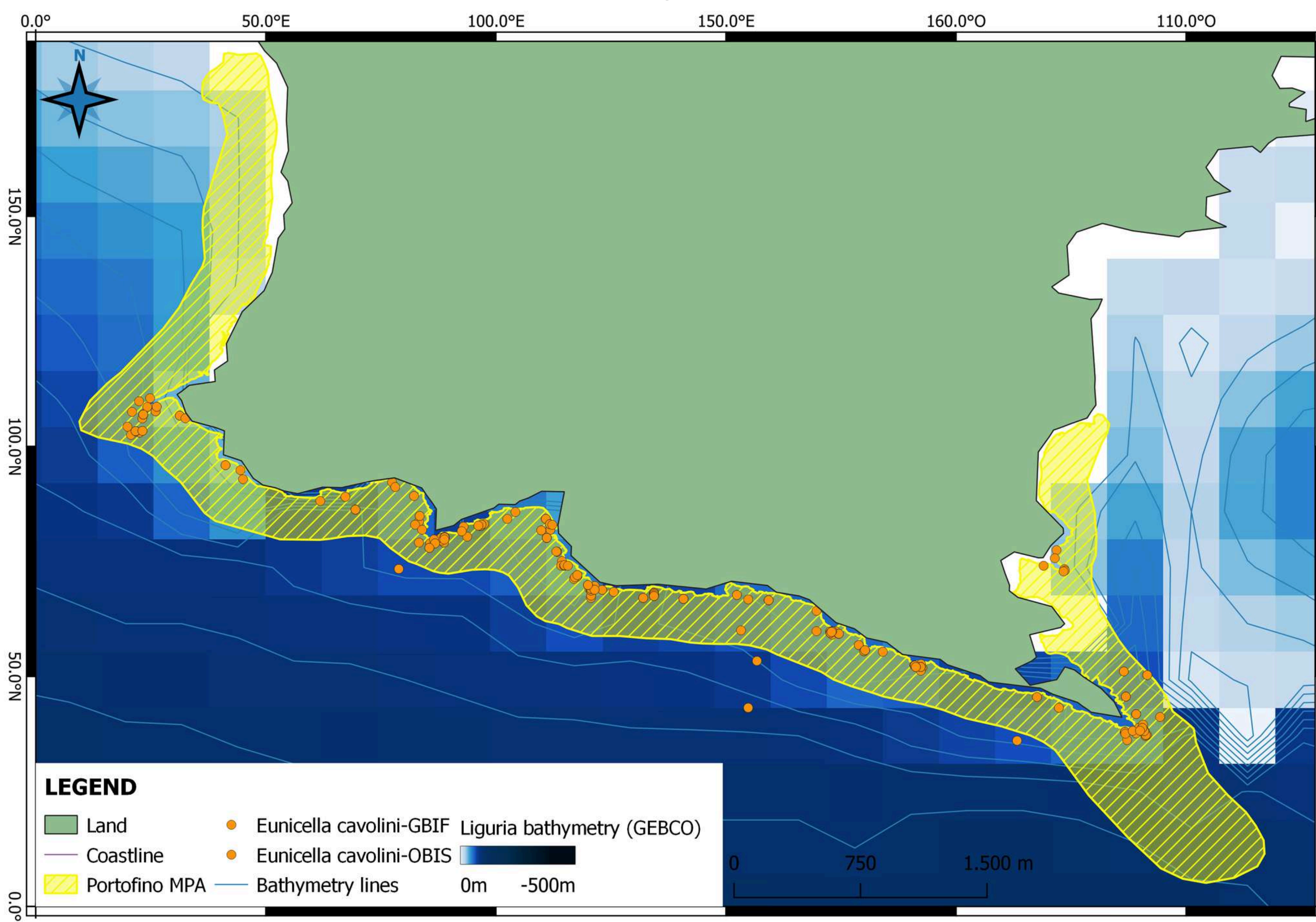
Natural disorders

- Climate anomalies
- Ocean acidification
- Exceptional storms
- Eutrophication and sedimentation



LOCATION

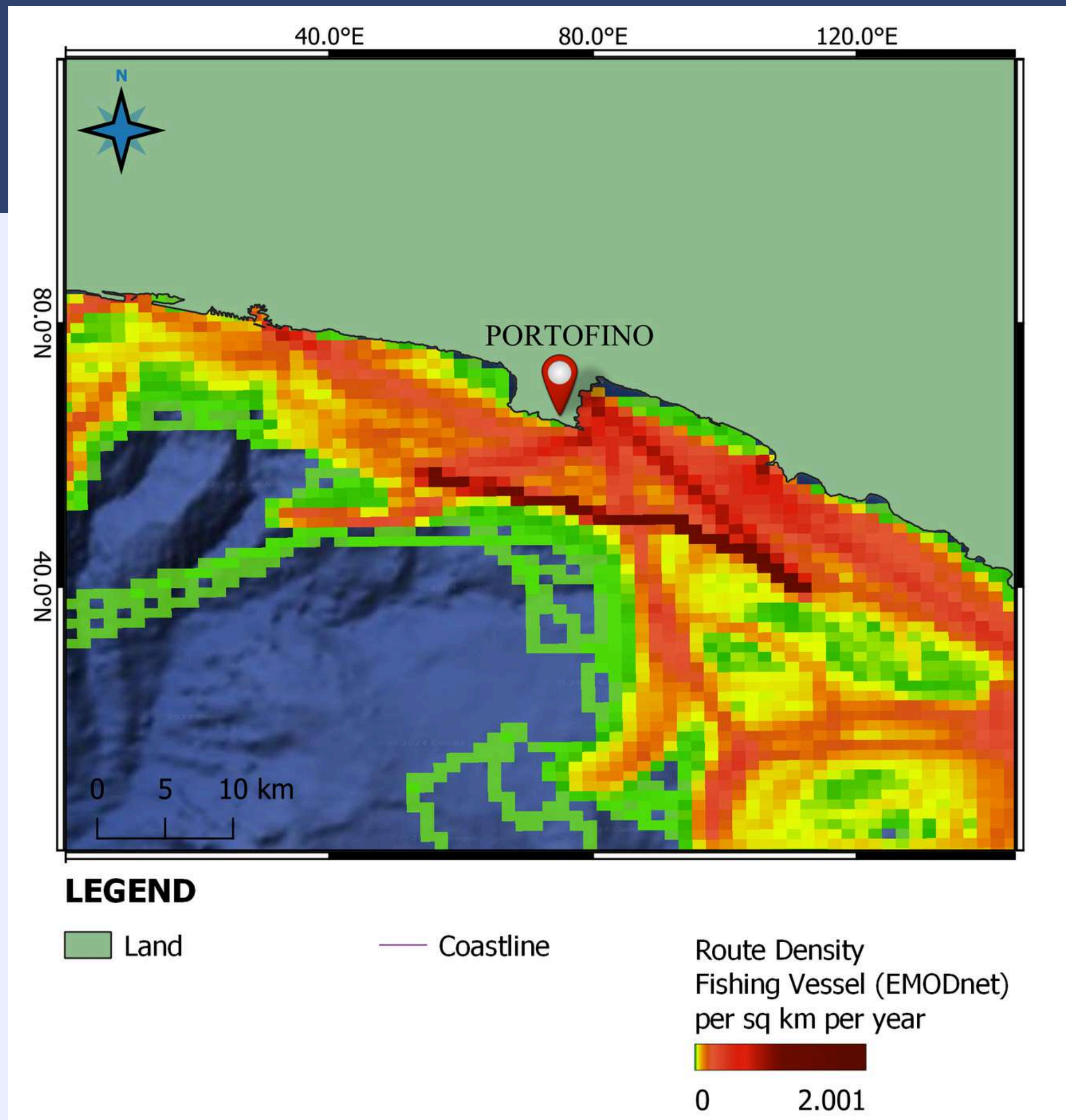




OCCURRENCES

OUR PROBLEM

- **Coastal Fishing :**
Can remove parts of the gorgonian forests
- **Effects inside MPA:**
~ 36% of the total *E. cavolinii* population is damaged by recreational / professional demersal fishing gears
- ~ 7% of the population severely damaged by entanglement



OUR CASE STUDY

- We **assume** a significant **damage event** , due to coastal fishing, occurred in 2022 east of Faro Cape
- On the border between MPA Zones B and C
- **Fewer restrictions** on fishing activities and **higher density of fishing vessel routes**



OUR GOAL



A LONG TERM
MONITORING STRATEGY
TO CHARACTERIZE AND
QUANTIFY THE BENTHIC
COMMUNITIES REPLACING
THE GORGONIAN FOREST

■ PHOTOSAMPLING

- Photos taken with 50x50 quadrats

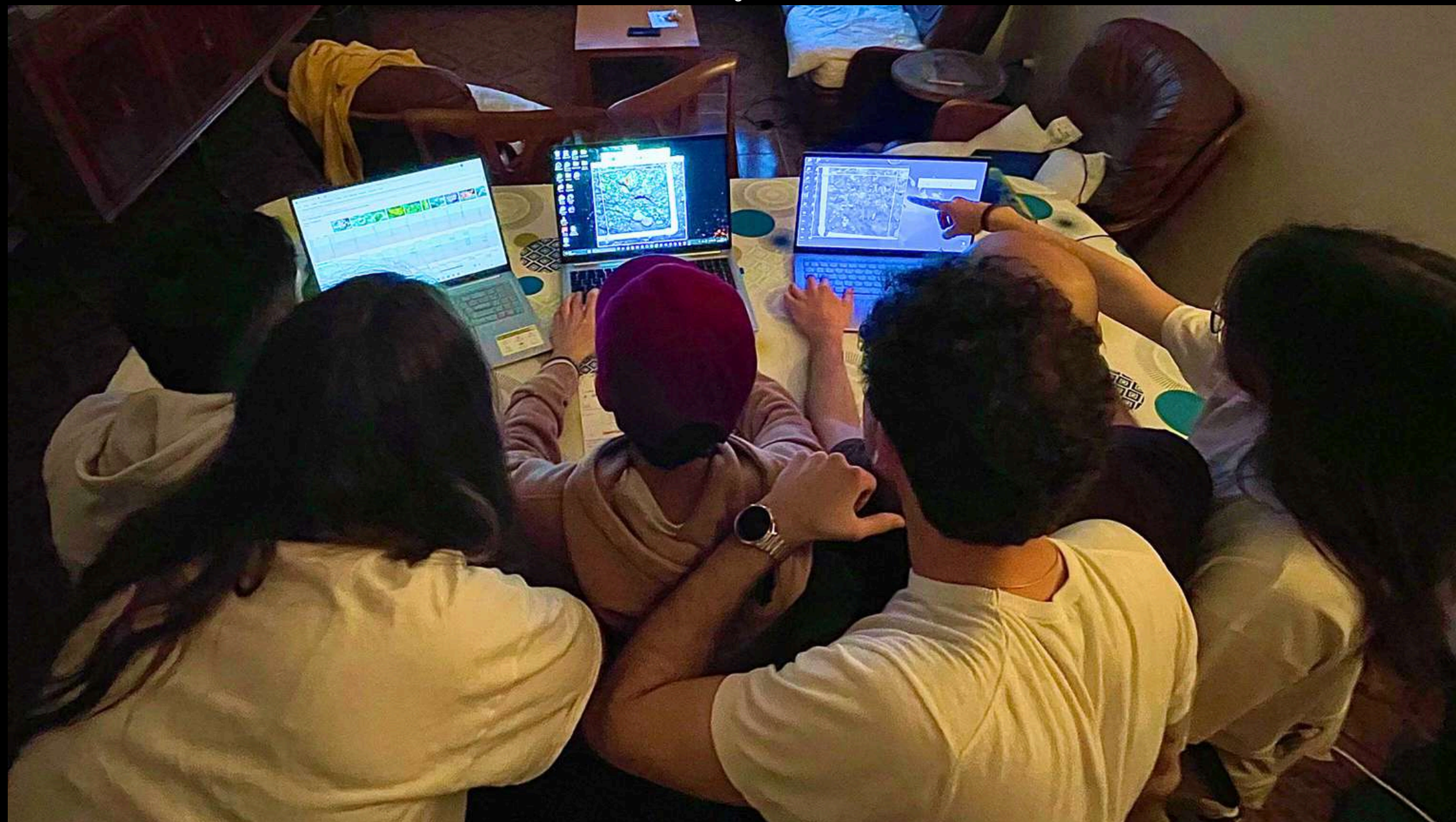
■ PERCENTAGE COVER

- Analyzing our photo-records with the **ImageJ** software, to assess the differences in percentage cover and species composition between impact and control sites
- Having a temporal comparison during the 5 years of sampling

■ 5 YEARS MONITORING



seaweed
Group



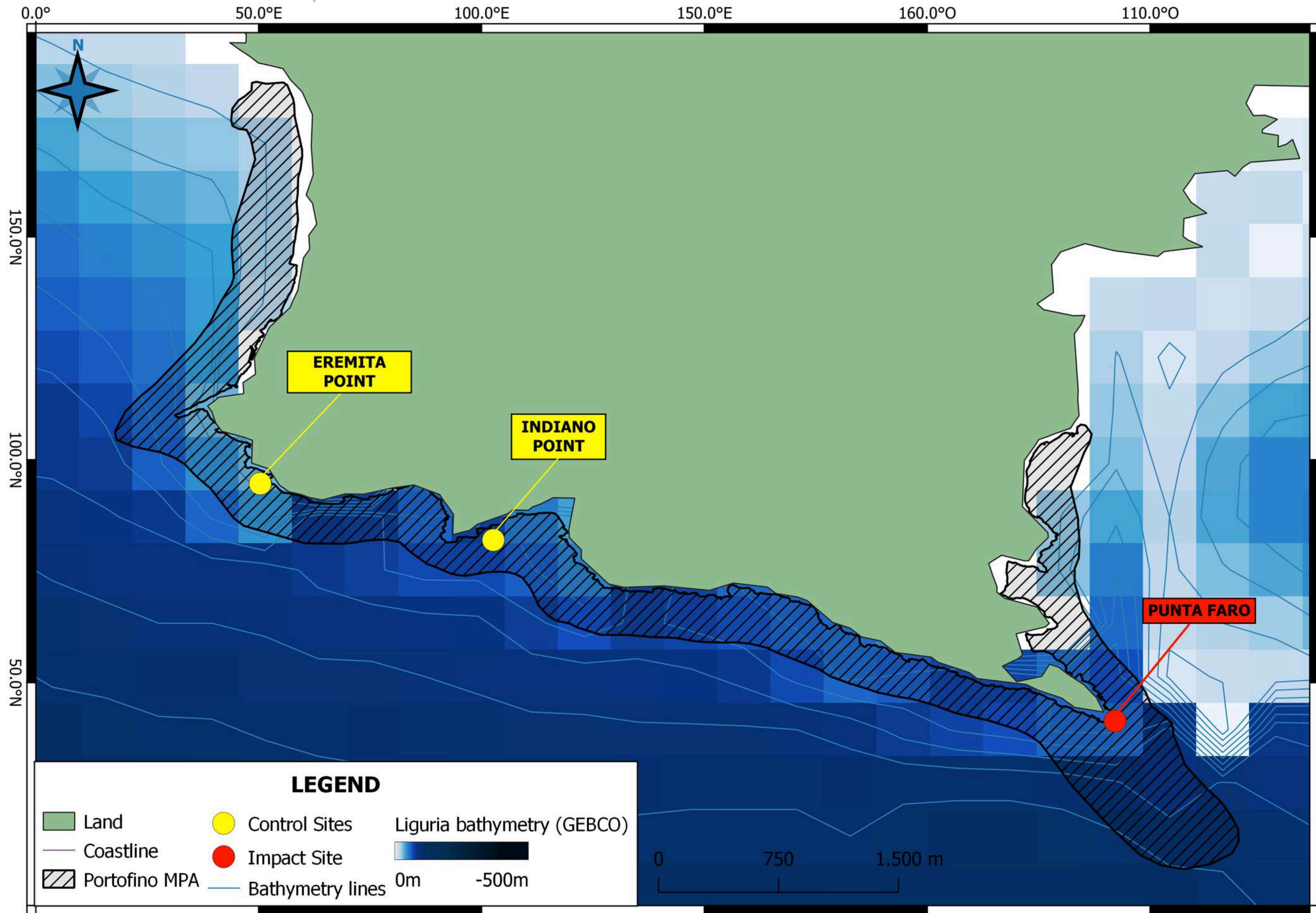
ImageJ
Image Processing and Analysis in Java



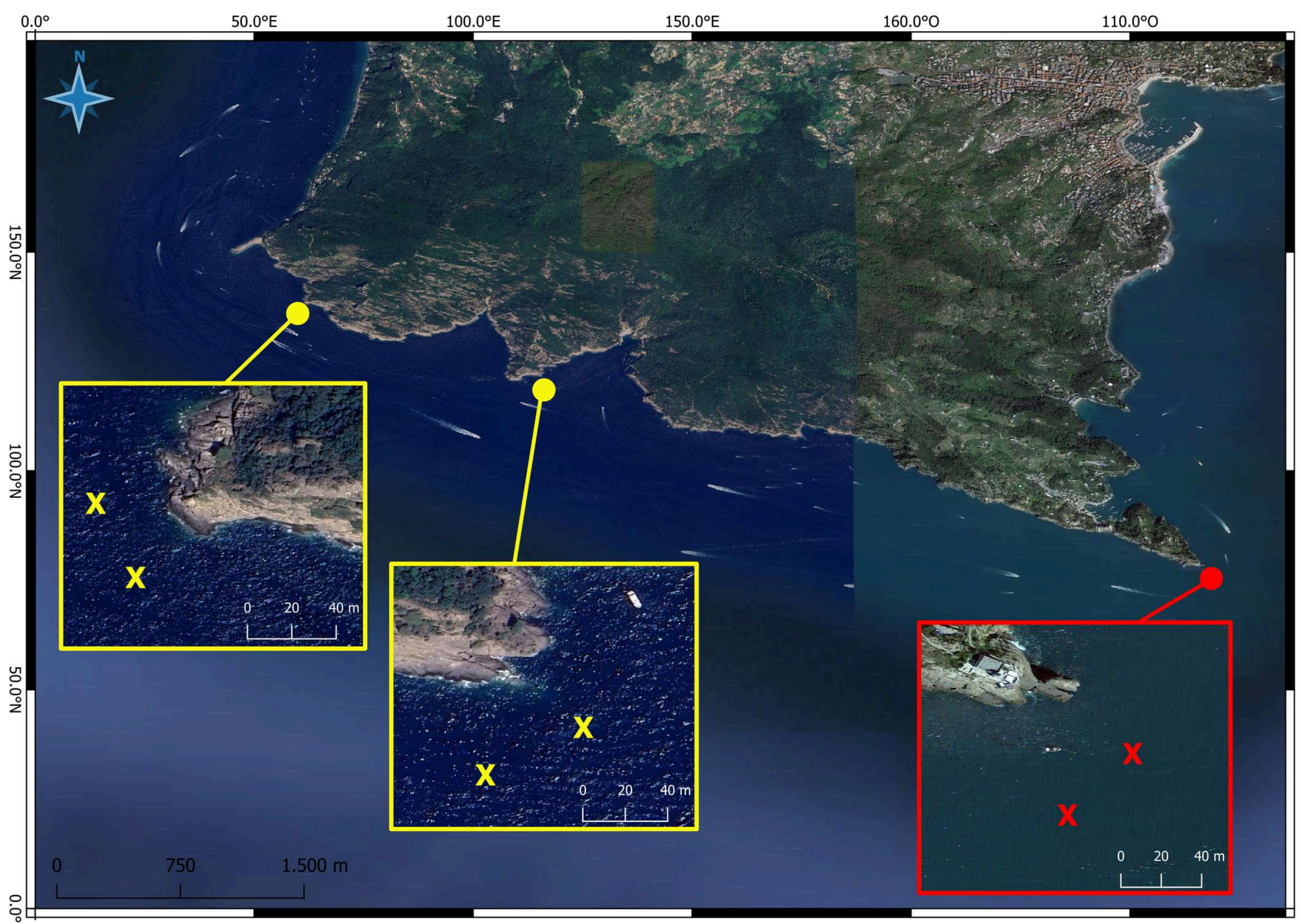
Seaweed group working on ImageJ

UNIVPM & UNIGE Marine Biology students could be involved in the ImageJ analysis process

SAMPLING AREA



SAMPLING AREA



SAMPLING DESIGN: NESTED STRATEGY

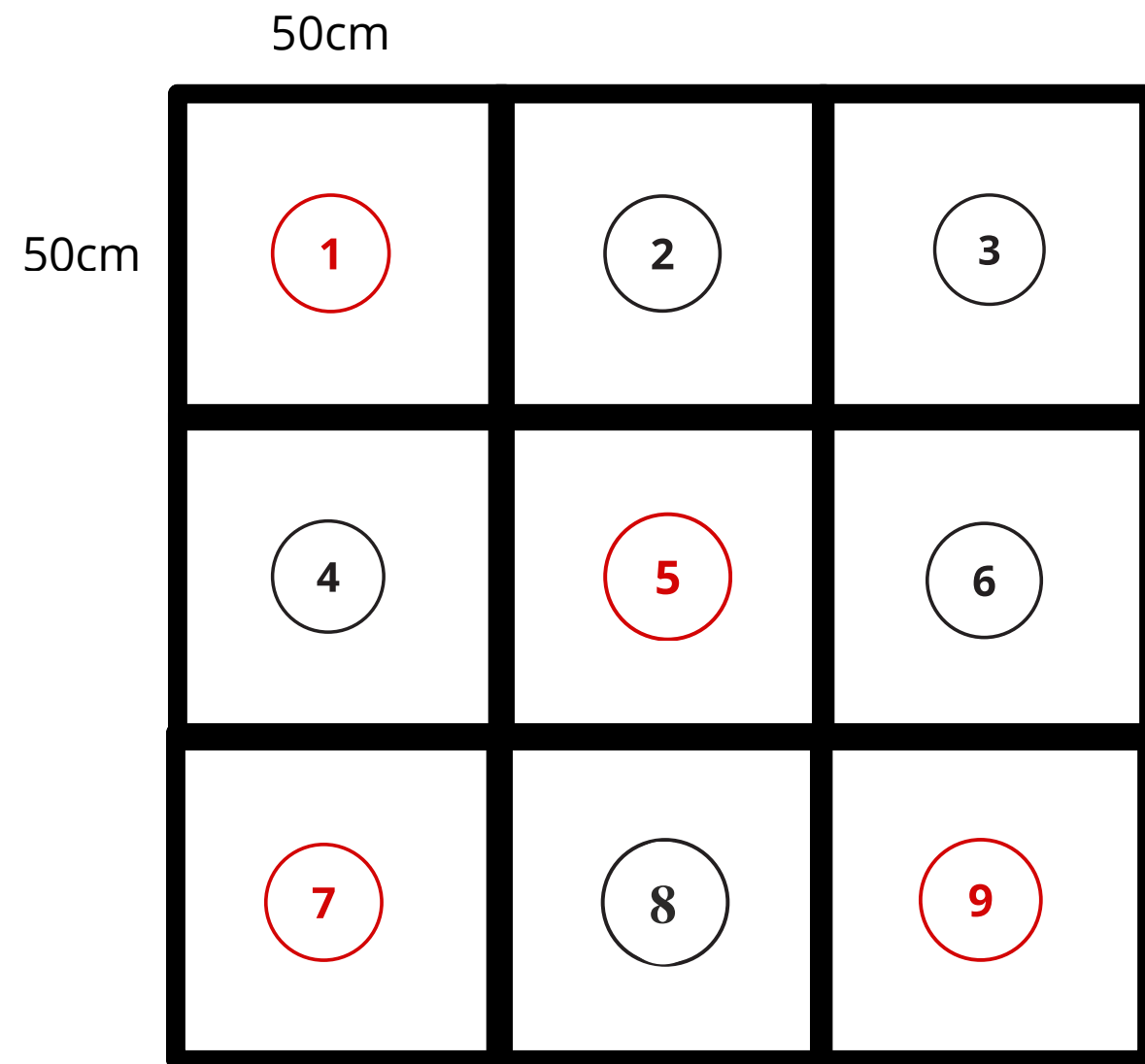
	IMPACT ZONE	CONTROL ZONE 1	CONTROL ZONE 2
AREA	1 2	1 2	1 2
REPLICATES	1 1 2 2 3 3 4 4	1 1 2 2 3 3 4 4	1 1 2 2 3 3 4 4

2 IMPACT ZONES
4 CONTROL ZONES



TOTAL = 24 SAMPLINGS
WITH 4 REPLICATES
FOR EACH AREA

PHOTOSAMPLING



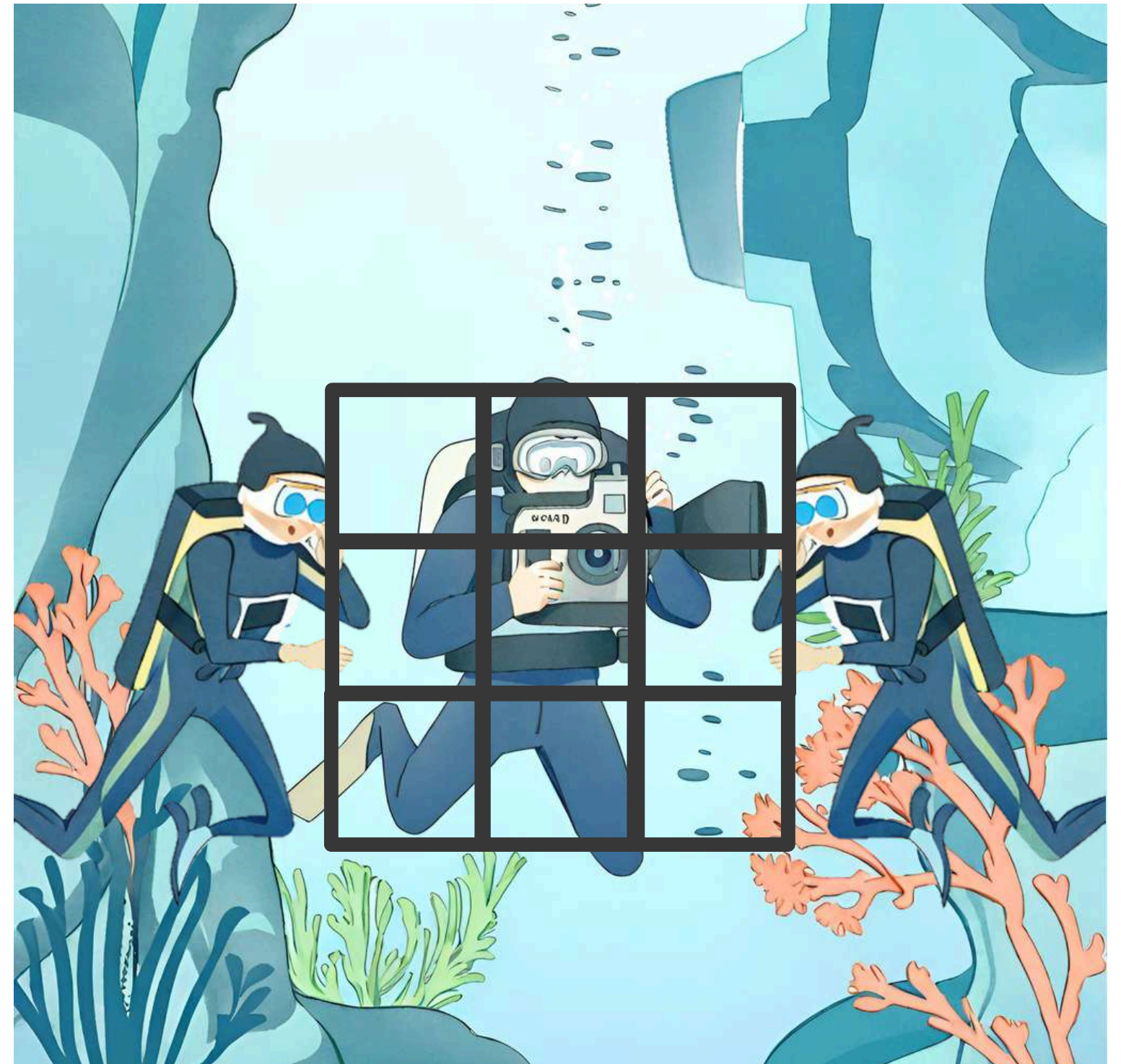
Ex. quadrats 1-5-7-9 drawn

- We build a **structure of 3x3 quadrats** (50cm x 50cm) to be taken to the different sampling sites
- Each quadrat is numbered from **1 to 9**
- By **Random Digit Table**, before each sampling-dive, 4 quadrats will be randomly drawn to be photosampled

4 random replicates for each area!

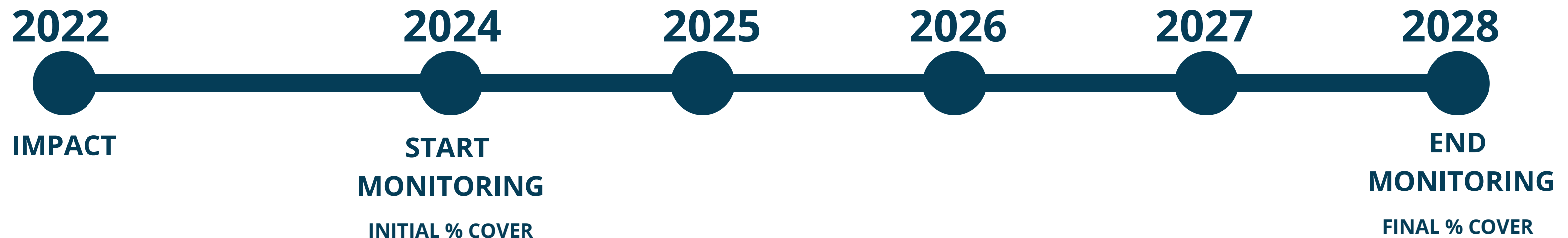
PHOTOSAMPLING

- The sampling is carried out by **3 divers**
- Once reached each site for the 1st time (2024), the precise **coordinates** and **depth** are noted by **GPS** and **Diving computer**
- 2 divers **hold** the structure **still** and **parallel** to the surface
- The 3rd diver **photosamples** the 4 quadrats previously drawn
- Finished the sampling they return to the surface



LONG-TERM MONITORING STRATEGY

- We want to monitor the **benthic communities replacing *E. cavolinii*'s forest** after the impact event
- We plan a **5-year monitoring strategy** assessing the percentage cover of the sampled areas by using the ImageJ software
- We want to monitor the **evolution of the percentage cover** of the different species during the 5 years, comparing also the initial percentage cover (**2024**) with the final one (**2028**)



MORE DETAILS ABOUT THE MONITORING STRATEGY

	STUDY AREA	CONSIDERED PARAMETERS	MONITORING
1st Photosampling 2 years after the impact event	Portofino 2 Impact areas 4 Control areas	Percentage cover Depth Turbidity Currents River runoffs Divers safety requirements	3 days between the end of August-September, 2 Sampling-dives per day: 1st day Impact area 2nd day control area 1 3rd day control area 2
5-years monitoring of the benthic communities replacing gorgonian forests	Portofino Same areas used for the 1st photosampling in 2024	Percentage cover Depth Turbidity Currents River runoffs Divers safety requirements	Once a year for 5 years Between the end of August-September

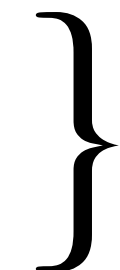
DIVE PLAN

ACTIVITY	TOOLS	DESIGN	TIME/REP	DEPTH	(15L AIR TANK) MAX BOTTOM TIME
Dive Photosampling of 4 random quadrats per area performed by 3 divers	3x3 quadrats structure (50x50cm) GPS Diving computer Compass UW camera	4 replicates per area, previously drawn	2'	27m	24min (1st dive)
					21min (2nd dive)

MORE DETAILS:

- v descent < 23 m/min;
- v ascent = 9 m/min;
- 4 replicates (2min each) = 8 min;
- Sampling set-up = 4 min
- **Total time for sampling area = 18 min < max bottom time**

1 sampling in the morning (**9:00**)
 1 sampling in the evening (**13:30**)
4 hours between the 2 daily dives



all these for 3 days

DIVE PLAN

1 - TABELLA DI DECOMPRESSIONE

Velocità di discesa: 23 m/min (max)

Velocità di risalita: 9 m/min

usare la tabella
soltanto con

ARIA

F.I.P.S.A.S.



6	9	12	15	18	21	24	27	30	33	36	39	42	45	prof.
		6	6	6	6	6	6	6	6	6	6	6	6	tappe
26	17	12	9	7	6	5	4	4	3	3	2	2		A
43	27	20	15	12	10	9	7	6	6	5	4	4	3	B
61	38	27	21	17	14	12	11	9	8	7	6	6	5	C
82	50	36	28	22	19	16	14	12	11	10	9	8	7	D
106	62	44	34	28	23	20	17	15	14	12	11	10	8	E
133	76	53	41	33	28	24	21	18	16	15	12		10	F
165	91	63	48	39	32	28	24	21	18	15	12			G
205	107	73	56	45	37	32	28	25	20	4		15	5	H
256	125	84	63	51	42	36	31		25	5				I
330	145	95	71	57	47	39	33	30	3			20	8	J
461	167	108	80	63	48	45	10					20	13	K
illim.	193	121	89	70	7	55	9							L
	223	135	92			60	14							M
	260	151	100	4	80	14								N
	307	163	110	8										O
	371	180	14											Z

ATTENZIONE! L'osservanza delle tabelle non esclude del tutto i rischi dell'immersione! Adottare comportamenti prudentiali

- Immergersi in perfette condizioni psicofisiche, in coppia, con attrezzatura completa ed efficiente, nei limiti di brevetto
- In presenza di fattori di rischio (stress, freddo, fatica, ecc.) usare il tempo di tabella successivo
- Raggiungere la profondità massima ad inizio immersione, non effettuare YO-YO
- Pianificare l'immersione in curva di sicurezza (consigliato raggiungere i 6 m entro la fine del tempo di tabella scelto)
- Sosta profonda (imm. in curva): 2,5 min a metà profondità max (stacco dal fondo 2,5 min prima del tempo tabellare)
- Sosta di sicurezza: 3 min a 6 m. Emersione: 3 m/min da 6 m alla superficie

consumi superiori a 2250 lt →
tempo entro la curva di sicurezza → 19
tempo limite in curva di sicurezza → 20
tempo fuori curva di sicurezza → 25 5
tempo di decompressione a 6 m →

NOTA: Per tempi e profondità intermedi a quelli in Tabella 1, 2 e 3 scegliere i valori immediatamente superiori

2 - INTERVALLO DI SUPERFICIE

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	Z
															0:10
															0:52
															0:10
															0:53
															1:45
															2:38
															3:30
															4:22
															5:14
															6:07
															7:51
															8:43
															9:35
															10:28
															11:19
															12:13
															13:06
															14:00
															15:50

3 - TEMPO DI AZOTO RESIDUO

F.A.R. iniziale
Profondità della immersione successiva

	6	9	12	15	18	21	24	27	30	33	36	39	42	45
Z	**													
O	**	+308	+169											
N	*	63	-											
M	**	+261	+152	+109										
L	*	110	11	-										
K	**	+224	+136	+99	+79									
J	*	147	27	-	-									
I	**	+194	+122	+90	+72	+59								
H	*	177	41	2	-	-								
G	+462	+168	+109	+81	+65	+54								
F	*	203	54	11	-	-								
E	+331	+146	+97	+73	+58	+49	+42	+7	+33	+30				
D	*	225	66	19	5	-	-	-	-	-				
C	+257	+126	+85	+65	+52	+44	+38	+3	+30	+27	+24			
B	*	245	78	27	11	4	1	-	-	-	-			
A	+206	+108	+74	+57	+46	+39	+33	+9	+26	+24	+22	+20	+19	
	*	263	89	35	17	9	6	-	-	-	-	-	-	-
	+166	+92	+64	+49	+40	+34	+29	+6	+23	+21	+19	+18	+16	+15
	*	279	99	43	23	14	10	2	-	-	-	-	-	-
	+134	+77	+55	+42	+35	+29	+25	+2	+20	+18	+17	+15	+14	+13
	*	294	108	50	28	19	14	11	5	2	-	-	-	-
	+106	+63	+45	+35	+29	+25	+22	+9	+17	+16	+14	+13	+12	+11
	*	308	118	57	34	23	17	14	8	4	1	-	-	-
	+83	+51	+37	+29	+24	+20	+18	+6	+14	+13	+12	+11	+10	+9
	*	320	126	63	39	28	21	17	11	7	3	1	-	-
	+62	+39	+29	+23	+19	+16	+14	+12	+11	+10	+9	+8	+7	+6
	*	332	134	69	44	32	25	21	14	10	6	3	2	-
	+44	+28	+21	+17	+14	+12	+10	+9	+8	+7	+6	+5	+4	+3
	*	343	142	75	49	36	29	24	17	12	8	6	4	2
	+27	+18	+13	+11	+9	+8	+7	+6	+5	+5	+4	+4	+4	+4
	*	353	150	81	54	40	32	27	20	15	10	8	6	4

- La reimmersione entro 10 min va considerata continuazione dell'immersione precedente
- Tra due immersioni successive si consiglia un intervallo di superficie minimo di 2 h
- L'immersione ripetitiva deve durare almeno 15 min ed essere meno profonda della precedente
- Non eccedere con le immersioni ripetitive per più giorni consecutivi ("settimane blu"). Interrompere la sequenza con adeguati periodi di riposo

F.A.R. finale
immersione non fattibile
tempo da aggiungere alla permanenza effettiva
tempo residuo in curva di sicurezza, (-) immersione fuori curva
intervallo oltre il quale l'immersione successiva non è ripetitiva

OUR HYPOTHESIS

Null hypothesis (H0)

- After the impact event on *E. cavolinii*, there will be **no** significant changes in the benthic communities

Alternative hypothesis (H1)

- After the impact event on *E. cavolinii*, there will be significant changes in the benthic communities

EXPECTED RESULTS

Ecological succession: what life comes after death

- **Without** gorgonians’ **shadow-effect**,
shift from a hard coralline dominance to a fleshy green algae one
(we expect to see that from the % cover and species composition analysis by ImageJ)
- Species abundances patterns change

ORGANISMS	<i>E. CAVOLINII</i> PRESENCE
Chlorophyta	N > Y
Phaeophyta	N > Y
Hydrozoans	Y > N
Polychaetes	N > Y

ORGANISMS	<i>E. CAVOLINII</i> PRESENCE
Foraminifera	N > Y
Bryozoa	Y > N
Encrusting sponges	N > Y



FUNDING

Italian Level:

- Marine Protected Area - Portofino
- The Italian Climate Fund - Ministero dell'Ambiente e della Sicurezza Energetica
- PNNR MER (Marine Ecosystem Restoration) - ISPRA



European Level:

- European Maritime, Fisheries and Aquaculture Fund (EMFAF)
- European Environment Agency

STAKEHOLDERS

- Marine Protected Area Portofino
- Divers
- Fishermen
- Italian Government (Ministero dell'Ambiente)
- European Environment Agency
- WWF Italy
- Università Politecnica Delle Marche
- Università di Genova



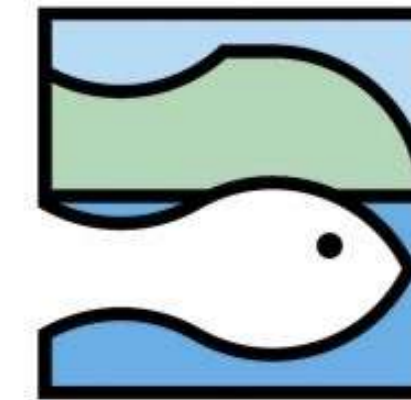
**Università
di Genova**



MINISTERO DELL'AMBIENTE
E DELLA TUTELA DEL TERRITORIO E DEL MARE



WWF



AREA MARINA PROTETTA

portofino



PADI



**European
Environment
Agency**



**UNIVERSITÀ
POLITECNICA
DELLE MARCHE**

LEGISLATIONS TO FOLLOW

- **Provisions for the defense of the sea**

(n. 979 of 31 December 1982)

- **Framework law on protected areas**

(n. 394 of 6 December 1991)

- **Official Journal no. 181 of 04/08/2008**

Regulates the nautical activity

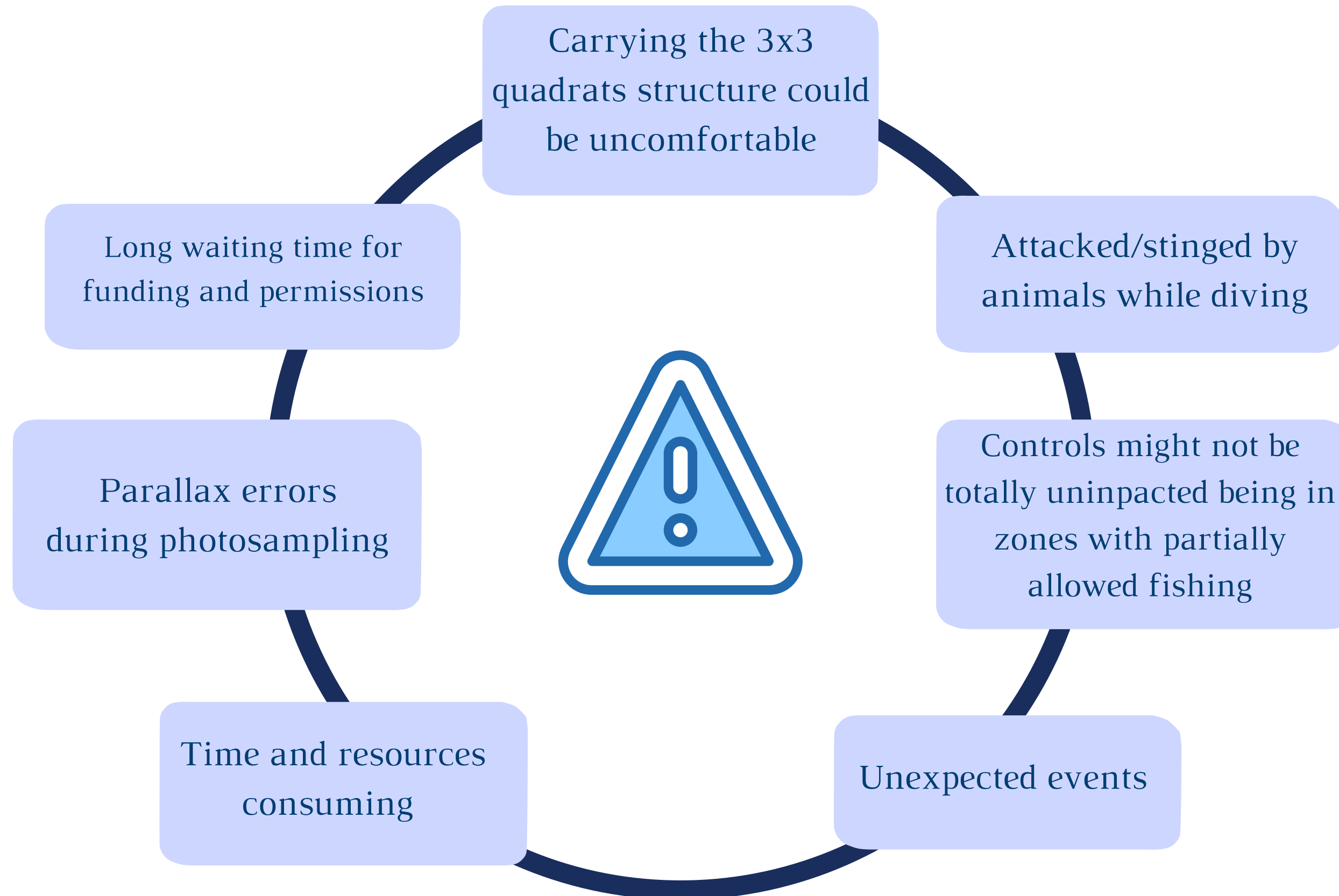


COSTS

Additional Costs	BOAT RENTAL	Personnel Scientific Salary	Travel Cost
	€6000	€ 4000	€ 900
	Cost per 3 days	3 Divers and Boat Crew	Travel, Hotel and Food
€ 3000	Tools		
€ 2000	Poor Weather Conditions		
€ 100 - 2000	Mechanical Damages		

- The total cost of each expedition is estimated to be around 10,900 euros.
- The total cost of the whole project (5 expditions) is estimated to be around 57,500 euros.

RISK ANALYSIS



POSSIBLE SOLUTIONS



Having professional
scientific divers in the
team to complete the task

Establish good
stakeholder
communication



Involve Universities
to divide tasks

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Thank you by Seaweed Group!

